# AHMED YASSER

AI & Computer Vision Engineer

Cairo, Egypt
+20 112 028 6999
ahmedyasser692004@gmail.com
GitHub: AhmedYasser06
LinkedIn: ahmed-yasser-ai

## PROFILE SUMMARY

AI & Machine Learning Engineer with a strong Computer Engineering foundation, specializing in Computer Vision, Natural Language Processing, and Large Language Models (LLMs). Skilled in Python, PyTorch, LangGraph, LangChain, SQL, and Docker, AWS with a strong track record in conducting research and driving cross-functional collaboration to deliver impactful AI solutions. Expertise in designing and deploying scalable model architectures to solve complex real-world challenges through predictive analytics and automation..

## CORE COMPETENCIES

| | | |
|---|---|---|
| **Machine Learning & Deep Learning** | **Embedded / Hardware Control** | **Deep Learning Research** |
| **LLM Agents & RAG Systems** | **Computer Vision Problems** | **Containers & Cloud Deployment** |

## PROFESSIONAL EXPERIENCE

### AI & Data Science Engineer — DEPI *Jun 2025 – Jan 2026*

- Deployed ML/DL models for predictive analytics, integrating them into automated production workflows.
- Engineered features across large-scale datasets to improve overall data quality for downstream models.
- Optimized model architectures (voting classifiers, neural networks) to boost predictive performance.
- Applied computer-vision and NLP techniques to deliver insight-driven, production-ready AI solutions.

## FLAGSHIP PROJECTS

### Multi-Layer Anti-Drone Defense System (GitHub)

*Python · YOLOv8 · Deep SORT · OpenCV · PID Control · Q-Learning (RL) · Arduino · Sensor Fusion*

- Built a real-time, sensor-fused counter-UAS platform: dual-model **YOLOv8** detection (RGB + thermal) running concurrently, geometrically fused via calibrated similarity-transform alignment for higher-confidence threat detection than either sensor alone.
- Designed a novel **NMSE-based threat-prioritization engine** that scores every live track on distance, closing speed, confidence, and size to decide which target to engage first.
- Implemented full **PID visual servoing** with anti-windup for gimbal control, plus an optional tabular **Q-learning agent** that adaptively tunes controller gains online while keeping engagement decisions deterministic and auditable.
- Split compute (laptop: inference, fusion, tracking, control) from actuation (Arduino: servos, radar sweep, jammer relay) — no Raspberry Pi required.

### CourseTA — Agentic Educational Content Platform

*LangChain · LangGraph · FastAPI · RAG · Whisper · PyMuPDF · Microservices*

- Architected a multi-agent educational system with **LangChain/LangGraph**, deploying specialized agents for question generation, summarization, and feedback-driven refinement.
- Built a **RAG-based QA pipeline** with embedding-based retrieval for fast, accurate knowledge access across course materials.
- Delivered real-time **async FastAPI** endpoints on a microservice architecture, integrating **Whisper** for audio/video transcription and **PyMuPDF** for PDF parsing across diverse content formats.
- Optimized the pipeline for human-in-the-loop workflows, enabling scalable, efficient content transformation.

### Relational Group Activity Recognition (ECCV 2018 reproduction) [(GitHub)]

*PyTorch · Graph Attention Networks (GAT) · CNN-LSTM*

- Re-implemented a Hierarchical Relational Network on **4,830 frames from 55 videos**, modeling inter-person relations and hierarchical temporal dynamics via a custom Graph Relational Layer — **91.55% test accuracy**.
- Added a **Graph Attention (GAT)** mechanism to sharpen relational modeling, pushing accuracy to **92.00% with test-time augmentation**.

### Group Activity Recognition (CVPR 2016 reproduction) [(GitHub)]

*PyTorch · Two-Stage LSTM · Temporal Modeling*

- Implemented a Hierarchical Temporal Deep Model with a two-stage LSTM architecture to capture multi-person temporal dynamics, reaching **93% accuracy** on benchmark data.
- Ran ablation studies isolating each component's contribution and benchmarked against **8 baseline models**, demonstrating state-of-the-art effectiveness.

### Multi-Task Chest X-Ray Analysis System [(GitHub)]

*PyTorch · DenseNet-121 · Multi-Task Learning · Medical Imaging*

- Designed **MultiCheXNet**, a multi-task DenseNet-121 model performing simultaneous classification, detection, and segmentation on chest X-rays through a shared encoder.
- Trained across multiple medical datasets with a custom dataloader to reconcile differing label schemas; shared representations improved both accuracy and inference speed over single-task baselines.
- Enabled modular weight sharing to support independent component pretraining as well as full end-to-end fine-tuning.

### Urban Scene Segmentation [(GitHub)]

*PyTorch · U-Net · DeepLabV3+ · ASPP · CamVid Dataset*

- Built and compared three progressively stronger segmentation baselines (U-Net → improved U-Net → DeepLabV3+ with ASPP) on the CamVid urban driving dataset across 12 semantic classes.
- Took mean IoU from **0.34 to 0.59** through Dice+CrossEntropy loss, mixed-precision training, and atrous multi-scale context aggregation, backed by full quantitative and boundary-level error analysis.

### AI-Powered Predictive Maintenance [(GitHub)]

*Python · MLOps · Docker · Time-Series Forecasting*

- Engineered a failure-forecasting system across **10+ sensors**, predicting equipment failures **24 hours in advance** — **84.7% recall**, **96% accuracy**, tuned to minimize false alerts.
- Deployed via containerized MLOps workflows for reliable, scalable operational performance.

### Credit Card Fraud Detection System [(GitHub)]

*PyTorch · Ensemble Learning · Focal Loss · Imbalanced Data*

- Tackled extreme class imbalance on **284,807 transactions** using a voting-classifier ensemble and a focal-loss-enhanced PyTorch neural network.
- Outperformed standard classifiers on rare-event detection: **86% F1**, **85% PR-AUC**.

## EDUCATION

### B.Sc. Computer and Systems Engineering — Faculty of Engineering, Zagazig University *2022 – 2027*

*Strong academic performance across core coursework, including:*

- **Computer Science Core:** Computer Programming, Data Structures, Algorithms, Logic Design, Computer Organization, Operating Systems, Database Systems, Computer Networks
- **AI & Systems:** Artificial Intelligence, Automatic Control, Digital Control, Process Control, Embedded Systems
- **Math & Engineering Foundations:** Engineering Mathematics, Statistics & Probability, Electric Circuits, Electronic Engineering

## COMPETITIONS & ACHIEVEMENTS

### Anti-Drone Innovation Competition — Team SkyGuard *ITC Egypt ADC 2026*

*6th International Telecom Conference (ITC Egypt), hosted by the Egyptian Air Defense College*

- Represented **Team Skyhawks**, building and presenting the **Multi-Layer Anti-Drone Defense System** as the team's competition entry.

- Ranked in the **Top 5 among 320 competing teams**. , advancing past an initial shortlist of **45 qualified teams**.

## TECHNICAL SKILLS

| | |
|---|---|
| **Programming** | Python, C, C++, Java, SQL, OOP, Data Structures & Algorithms |
| **Deep Learning** | CNNs, RNNs/LSTMs, GNNs, GAT, VAE, Transformers, DETR, Faster-RCNN, YOLO |
| **AI / LLM Engineering** | PyTorch, LangChain, LangGraph, RAG, FastAPI, OpenCV |
| **Systems & Deployment** | Docker, Kubernetes, AWS, Azure, Git/GitHub, MLOps |
| **Applied Knowledge** | Signal Processing, Image Processing, Parallel Processing, Databases |

## CERTIFICATIONS & TRAINING

### DIPLOMAS & PROGRAMS

- Deep Learning and System Design Diploma — CSkilled
- Machine Learning Diploma — CSkilled
- Microsoft Machine Learning — DEPI

### SUMMER TRAINING

- Generative AI Summer Training — ITI
- AI Summer Training — ITI

### LANGUAGES

- Arabic : Native
- English : Excellent